

EMAIL SECURITY ANALYSIS & PHISHING DETECTION PIPELINE

A Cybersecurity Email Analysis and SIEM Monitoring Project

Project Documentation

Project Type: Cybersecurity / Email Security

Platform: Windows

SIEM: Splunk Enterprise

Email Platform: Gmail

Analysis Method: Rule-Based Email & Security Indicator Analysis

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PROJECT OVERVIEW & OBJECTIVES

1. Project Overview

The Email Security Analysis & Phishing Detection Pipeline is a cybersecurity project designed to collect, analyze, and monitor email security indicators from Gmail.

The system uses a Python-based Gmail IMAP collector to monitor incoming emails from both the INBOX and SPAM folders. Collected emails are parsed to extract important information such as sender, recipient, subject, message body, URLs, domains, and email headers.

The extracted information is then analyzed using SPF, DKIM, and DMARC authentication results along with suspicious subject and URL indicators. Selected URLs can also be analyzed using VirusTotal for additional threat-intelligence information.

The processed email data is forwarded to Splunk Enterprise, where it becomes searchable and is presented through dashboards for security monitoring and investigation.

Overall Workflow

Email Generation → Gmail → IMAP Collection → Email Parsing → Authentication Analysis → URL/Domain Analysis → VirusTotal → Splunk → Detection & Visualization

2. Project Objectives

The main objectives of this project are:

- To develop a Gmail IMAP-based email collection system.
- To monitor newly received emails from INBOX and SPAM.
- To extract useful email metadata, headers, URLs, and domains.
- To analyze SPF, DKIM, and DMARC authentication results.
- To identify potentially suspicious emails using multiple indicators.
- To perform URL reputation analysis using VirusTotal.
- To send structured email events to Splunk Enterprise.
- To create a centralized dashboard for email-security monitoring.
- To provide security analysts with searchable data for investigation.

- To demonstrate an end-to-end email threat-analysis workflow in a controlled laboratory environment.

SYSTEM ARCHITECTURE AND WORKFLOW

3. System Architecture

The Email Security Analysis & Phishing Detection Pipeline consists of multiple components that work together to collect, process, analyze, enrich, and visualize email security data.

The system follows an end-to-end workflow beginning with controlled email generation and ending with security analysis and visualization in Splunk Enterprise.

3.1 System Workflow

1. Controlled Email Generation

A custom Gmail Mail Sender application is used to create controlled test emails with different subjects, message bodies, URLs, and attachments.

2. Gmail Mailbox

The generated emails are delivered to a Gmail account where they can appear in the INBOX or SPAM folder.

3. Gmail IMAP Collector

The custom Python-based IMAP Email Collector connects to Gmail and monitors both INBOX and SPAM folders for newly received messages.

4. Email Parsing

The collector extracts important email information including sender, recipient, subject, body, URLs, domains, attachments, and email headers.

5. Authentication Analysis

SPF, DKIM, and DMARC authentication results are extracted from the email headers and converted into structured fields for further analysis.

6. Suspicious Email Detection

Email characteristics such as suspicious subjects, URL presence, domain information, and authentication results are evaluated to identify potentially suspicious messages.

7. URL Reputation Analysis

Extracted URLs can be submitted to VirusTotal to obtain additional threat-intelligence and reputation information.

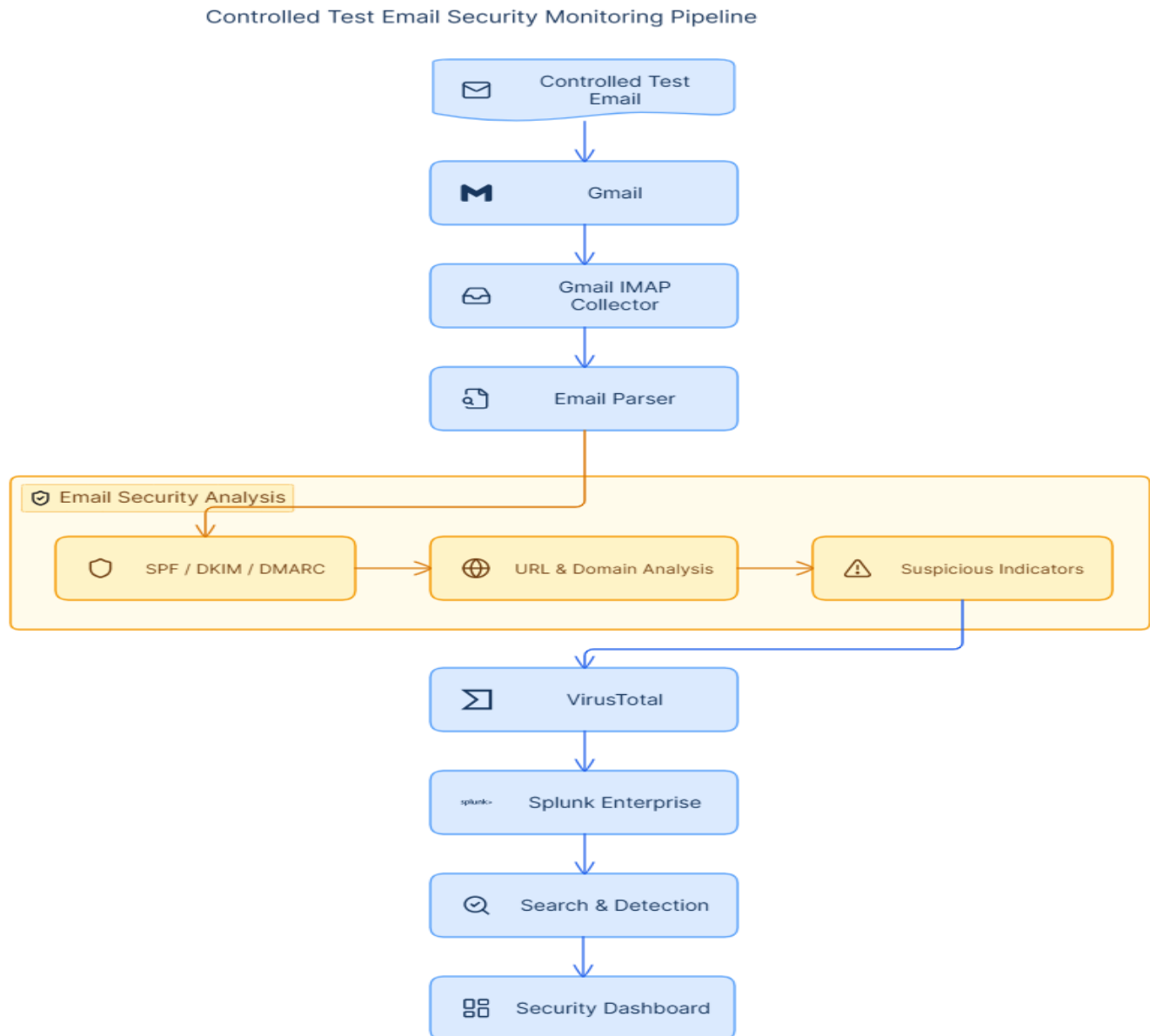
8. Splunk SIEM Integration

The processed email information is stored as structured events and ingested into Splunk Enterprise for centralized searching, analysis, and correlation.

9. Dashboard and Visualization

Splunk dashboards present email volume, suspicious email counts, authentication results, URLs, domains, subjects, senders, and other security-related information.

3.2 High-Level Architecture



3.3 Data Flow

The system converts an email message from an unstructured communication into structured security information. The collected data is parsed into individual fields and enriched with authentication and URL-reputation information. The resulting structured events are then indexed in Splunk, allowing the analyst to search and visualize the collected security indicators.

This architecture provides a practical workflow for investigating suspicious email activity within a controlled cybersecurity laboratory environment.

TOOLS AND TECHNOLOGIES USED

4. Tools and Technologies Used

The project combines programming, email-analysis, threat-intelligence, and SIEM technologies to implement an end-to-end email security analysis workflow.

4.1 Python

Python is the primary programming language used to develop the email collection, parsing, analysis, and supporting components of the project.

Python is used for Gmail IMAP communication, email extraction, header processing, URL and domain extraction, security-indicator analysis, and preparation of structured data for Splunk.

4.2 Gmail IMAP Collector

Gmail IMAP is used to retrieve email messages from the test mailbox.

The custom IMAP collector monitors the mailbox and retrieves newly received messages from both the INBOX and SPAM folders.

4.3 Email Parsing and Header Analysis

The project parses received emails to extract important information including:

- Sender
- Recipient
- Subject
- Message body
- Email headers
- URLs
- Domains
- Attachments
- Authentication results

4.4 MXToolbox

MXToolbox is used for detailed email header and domain-related analysis.

The project uses MXToolbox to examine email headers and inspect authentication-related information such as:

- SPF
- DKIM
- DMARC
- Authentication-Results

- Received headers
- Return-Path
- Message routing information

MXToolbox provides an additional analysis interface for validating and understanding the technical information contained within email headers.

4.5 SPF

Sender Policy Framework (SPF) is used to determine whether the sending server is authorized to send email on behalf of a domain.

The SPF result is extracted from email authentication information and used as a security indicator during analysis.

4.6 DKIM

DomainKeys Identified Mail (DKIM) provides cryptographic authentication for email messages.

The project extracts DKIM-related information from email headers and uses the authentication result as part of the email-security analysis.

4.7 DMARC

Domain-based Message Authentication, Reporting and Conformance (DMARC) provides a policy and authentication framework based on SPF and DKIM.

DMARC information is analyzed together with SPF and DKIM results to provide additional context during email investigation.

4.8 VirusTotal

VirusTotal is used as an external threat-intelligence source for URL reputation analysis.

URLs extracted from emails can be analyzed using VirusTotal to identify detections reported by multiple security vendors.

4.9 Splunk Enterprise

Splunk Enterprise is used as the central SIEM platform for the project.

Structured email events are ingested into Splunk, where they can be searched, filtered, analyzed, correlated, and visualized through dashboards.

4.10 Custom Gmail Mail Sender

A custom Gmail Mail Sender application is used to generate controlled test emails.

The application supports different test scenarios and allows the generation of messages containing specific subjects, content, URLs, recipients, CC/BCC values, and attachments.

4.11 Overall Technology Stack

Programming Language:

Python

Email Platform:

Gmail

Email Protocol:

IMAP

Email Header Analysis:

MXToolbox

Email Authentication:

SPF, DKIM, DMARC

Threat Intelligence:

VirusTotal

SIEM:

Splunk Enterprise

Testing:

Custom Gmail Mail Sender

These technologies work together to provide a complete workflow for collecting, analyzing, enriching, detecting, and visualizing email-security events.

CONTROLLED EMAIL GENERATION

5. Controlled Email Generation

The first stage of the project involves generating controlled test emails that can be analyzed by the email security pipeline. A custom Gmail Mail Sender application was developed to create different email scenarios and deliver them to the designated Gmail analysis account.

5.1 Gmail Mail Sender

The Gmail Mail Sender provides a graphical interface for configuring and sending test emails. It allows different email parameters to be controlled before sending the message.

The application provides options for:

- Recipient configuration
- CC and BCC configuration
- Subject selection
- Email body and custom content
- Predefined email types
- URL-containing messages
- Attachment support
- Multiple email generation
- Configurable sending delay

5.2 Test Email Scenario

For the demonstrated test, a security-related email containing a password-reset style message and a URL was generated.

The purpose of using controlled test messages is to create known email samples that can be passed through the complete analysis pipeline. This allows the collection, parsing, authentication analysis, URL extraction, and Splunk monitoring components to be tested using repeatable inputs.

5.3 Role in the Security Pipeline

The generated email acts as the initial input to the project.

The workflow begins as follows:

```
Gmail Mail Sender
↓
Test Email
↓
Gmail Mailbox
↓
IMAP Email Collector
↓
Email Analysis Pipeline
```

The generated messages are not treated as automatically malicious. They are controlled test inputs used to evaluate how the security-analysis system identifies and processes different email indicators.

The screenshot shows a web application titled "Email Sender" with a light gray background. The interface is divided into several sections:

- Mail Configuration:** Contains input fields for "Recipient" (with a placeholder "@gmail.com"), "CC", "BCC", "Subject" (with the text "password reset request"), "Mail Count" (set to 1), and "Delay (seconds)" (set to 2).
- Mail Type:** Features a dropdown menu labeled "Select Mail Type" with "Custom" selected. A list of options is visible: Account Notification, Password Reset, Invoice, Order Confirmation, Delivery Notification, Security Notification, HR Notification, Subscription Renewal, Newsletter, and Custom (highlighted in blue).
- Message:** Displays a preview of an email. It starts with "Hello," followed by a paragraph: "We received a request to reset the password associated with your account." Below this is a link: "If you made this request, continue here: https://[redacted]". Another paragraph follows: "If you did not request a password reset, you can safely ignore this message." The email ends with "Regards, Account Support".
- Attachments:** Includes buttons for "Add Files" and "Clear", and a text input field containing "invoice.pdf.txt".
- SEND EMAILS:** A prominent button at the bottom left of the interface.
- Ready:** A status indicator at the bottom right.

The application is running on a Windows operating system, as evidenced by the taskbar at the bottom showing the Start button, a search bar, and various application icons including File Explorer, Microsoft Edge, and Google Chrome.

Figure 5.1 — Custom Gmail Mail Sender interface used to generate controlled test emails.

TEST EMAIL RECEIVED IN GMAIL

6. Test Email Received in Gmail

After generating the controlled test email, the message is delivered to the designated Gmail mailbox. This stage confirms that the generated email has successfully entered the environment where it will be collected and analyzed.

6.1 Email Delivery

The received message contains the subject, sender information, message body, and URL that were configured during the email-generation stage.

The received email serves as the original source message for the subsequent analysis process.

6.2 Role in the Pipeline

The Gmail mailbox acts as the intermediate layer between the email-generation component and the IMAP collection component.

The workflow at this stage is:

Custom Gmail Mail Sender

↓

Gmail Mailbox

↓

INBOX / SPAM

↓

Gmail IMAP Collector

Once the message is available in the mailbox, the IMAP collector can detect and retrieve it for further processing.

6.3 Purpose of This Stage

Verifying the received email is important because it confirms that the test message has reached the monitored mailbox before the collection and analysis stages begin.

The original message is later correlated with its parsed data, email headers, authentication information, URLs, and Splunk event.

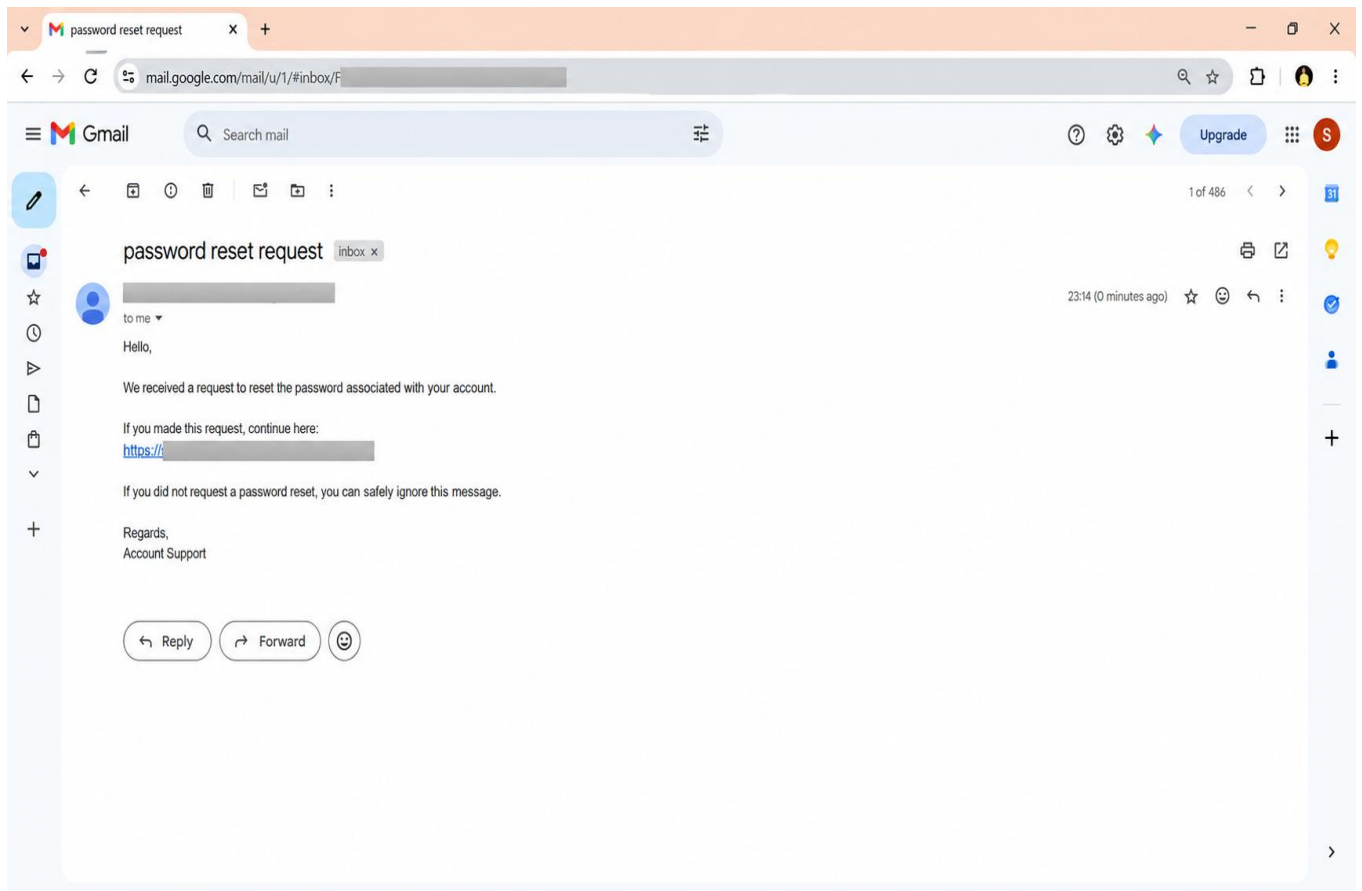


Figure 6.1 — Controlled test email received in the Gmail mailbox.

EMAIL IMAP EMAIL COLLECTOR

7. Gmail IMAP Email Collector

The Gmail IMAP Email Collector is responsible for monitoring the designated Gmail mailbox and collecting newly received email messages for security analysis.

The collector establishes an IMAP connection with Gmail and continuously checks the configured mail folders for new messages.

7.1 Mailbox Monitoring

The collector is configured to monitor both:

- INBOX
- SPAM

Monitoring both folders is useful for email-security analysis because suspicious or phishing-related messages may be automatically classified as spam by the mail provider.

7.2 Email Collection

When a new message is detected, the collector retrieves the email and processes it for further analysis.

The collection process provides information that can be used by the parsing and analysis components, including:

- Sender information
- Recipient information
- Subject
- Email content
- Email headers
- URLs
- Domains
- Attachments
- Authentication-related information

7.3 URL Extraction

The collector also identifies URLs contained within received messages.

The number of extracted URLs is displayed in the collector interface, providing an initial indicator that can later be used for URL and threat-intelligence analysis.

7.4 Role in the Security Pipeline

The IMAP collector connects the Gmail mailbox with the email-analysis components.

The workflow is:

Gmail Mailbox

↓

IMAP Connection

↓

INBOX + SPAM Monitoring

↓

New Email Detection

↓

Email Retrieval

↓

Email Parsing & Security Analysis

This component forms the bridge between the received email and the automated security-analysis pipeline.

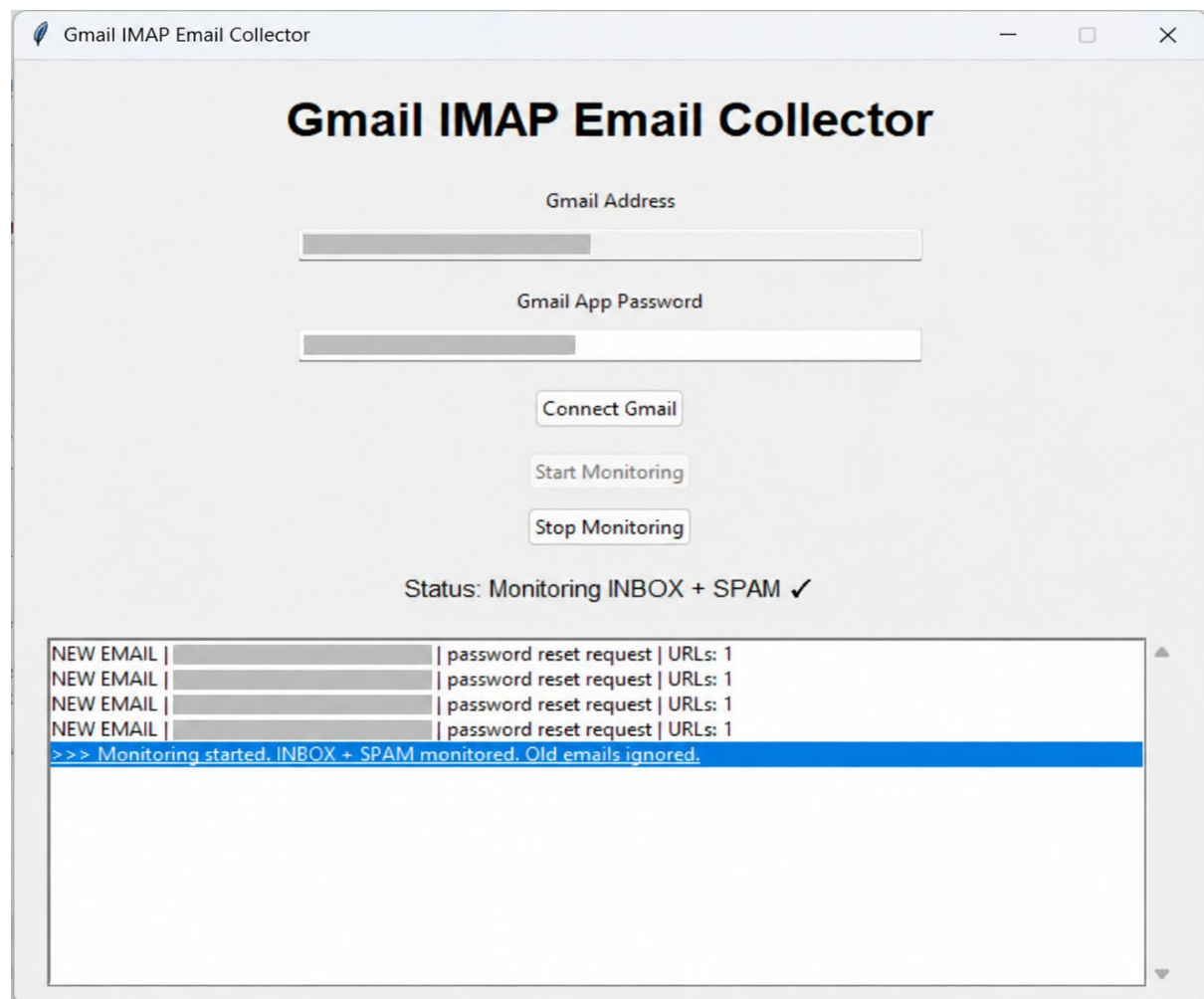


Figure 7.1 — Gmail IMAP Email Collector monitoring the INBOX and SPAM folders and detecting newly received emails.

EMAIL HEADER ANALYSIS USING MXTOOLBOX

8. Email Header Analysis Using MXToolbox

After collecting the email, the complete email header is analyzed using MXToolbox. Email headers contain technical information about the sender, delivery path, authentication mechanisms, and message processing history.

8.1 Header Analysis

The email header contains several important fields that are useful during security investigation, including:

- From
- To
- Date
- Subject
- Return-Path
- Received
- Received-SPF
- Authentication-Results
- DKIM-Signature
- ARC headers
- Message-ID
- Mail routing information

8.2 Authentication Information

MXToolbox is used to inspect authentication-related information present in the email header.

The analysis focuses on SPF, DKIM, and DMARC-related results because these mechanisms provide important information about email authentication and domain authorization.

8.3 Header Routing Information

The Received headers are also examined to understand the path followed by the email during delivery.

This information can help an analyst investigate the sending infrastructure and identify inconsistencies or unusual routing information.

8.4 Role of MXToolbox

MXToolbox acts as a supporting analysis tool in the project. It provides a readable representation of the technical information contained within the raw email header.

The analysis process is:

Collected Email



Email Header



MXToolbox

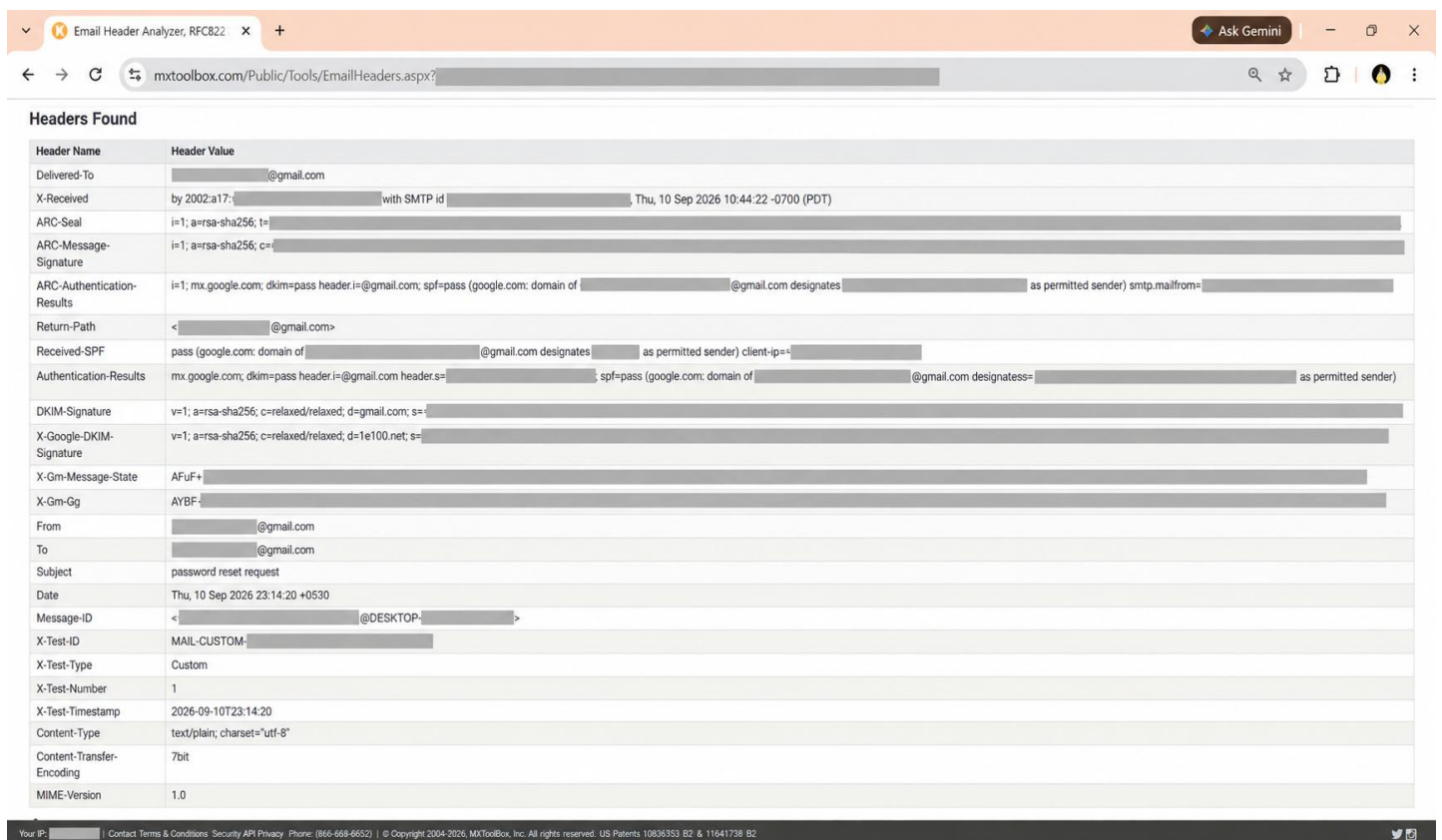


Header & Authentication Analysis



Security Investigation

The following three screenshots document different parts of the MXToolbox analysis.



The screenshot displays the MXToolbox Email Header Analyzer interface. The browser address bar shows the URL `mxtoolbox.com/Public/Tools/EmailHeaders.aspx?`. The page title is "Email Header Analyzer, RFC822". The main content area is titled "Headers Found" and contains a table with two columns: "Header Name" and "Header Value". The table lists various email headers and their corresponding values, including delivery information, authentication results, and message details. The footer of the page contains a dark bar with the text "Your IP: [redacted] | Contact Terms & Conditions Security API Privacy Phone: (866-668-6652) | © Copyright 2004-2026, MXToolBox, Inc. All rights reserved. US Patents 10836353 B2 & 11641738 B2" and social media icons for Twitter and Facebook.

Header Name	Header Value
Delivered-To	[redacted]@gmail.com
X-Received	by 2002a17-[redacted] with SMTP id [redacted], Thu, 10 Sep 2026 10:44:22 -0700 (PDT)
ARC-Seal	i=1; a=rsa-sha256; t=[redacted]
ARC-Message-Signature	i=1; a=rsa-sha256; c=[redacted]
ARC-Authentication-Results	i=1; mx.google.com; dkim=pass header.i=@gmail.com; spf=pass (google.com: domain of [redacted]@gmail.com designates [redacted] as permitted sender) smtp.mailfrom=[redacted]
Return-Path	<[redacted]@gmail.com>
Received-SPF	pass (google.com: domain of [redacted]@gmail.com designates [redacted] as permitted sender) client-ip=[redacted]
Authentication-Results	mx.google.com; dkim=pass header.i=@gmail.com header.s=[redacted]; spf=pass (google.com: domain of [redacted]@gmail.com designates [redacted] as permitted sender)
DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=gmail.com; s=[redacted]
X-Google-DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=1e100.net; s=[redacted]
X-Gm-Message-State	AFuF+[redacted]
X-Gm-Gg	AYBF-[redacted]
From	[redacted]@gmail.com
To	[redacted]@gmail.com
Subject	password reset request
Date	Thu, 10 Sep 2026 23:14:20 +0530
Message-ID	<[redacted]@DESKTOP-[redacted]>
X-Test-ID	MAIL-CUSTOM-[redacted]
X-Test-Type	Custom
X-Test-Number	1
X-Test-Timestamp	2026-09-10T23:14:20
Content-Type	text/plain; charset="utf-8"
Content-Transfer-Encoding	7bit
MIME-Version	1.0

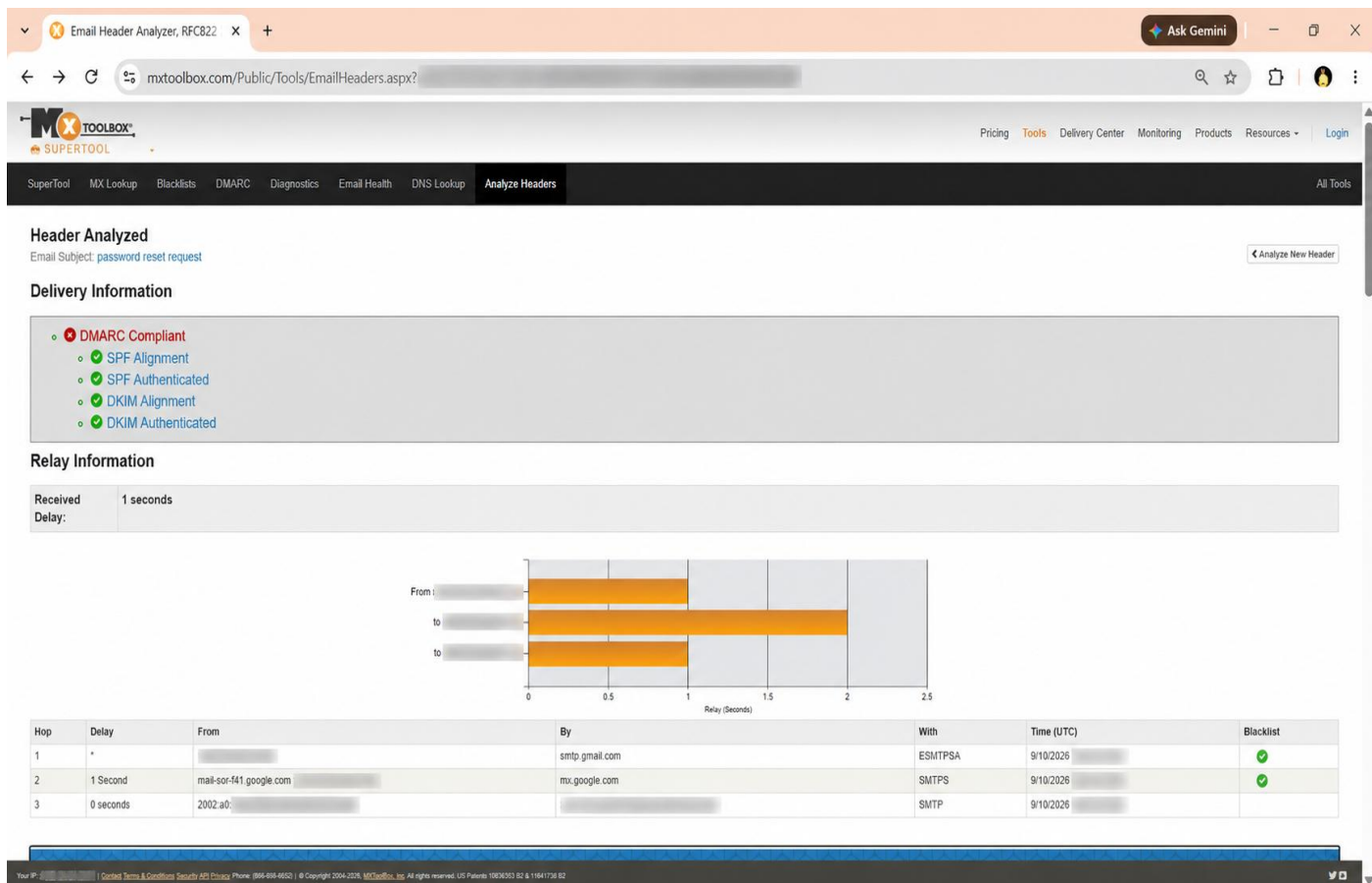


Figure 8.2 — Authentication and domain related information analyzed using MXToolbox.

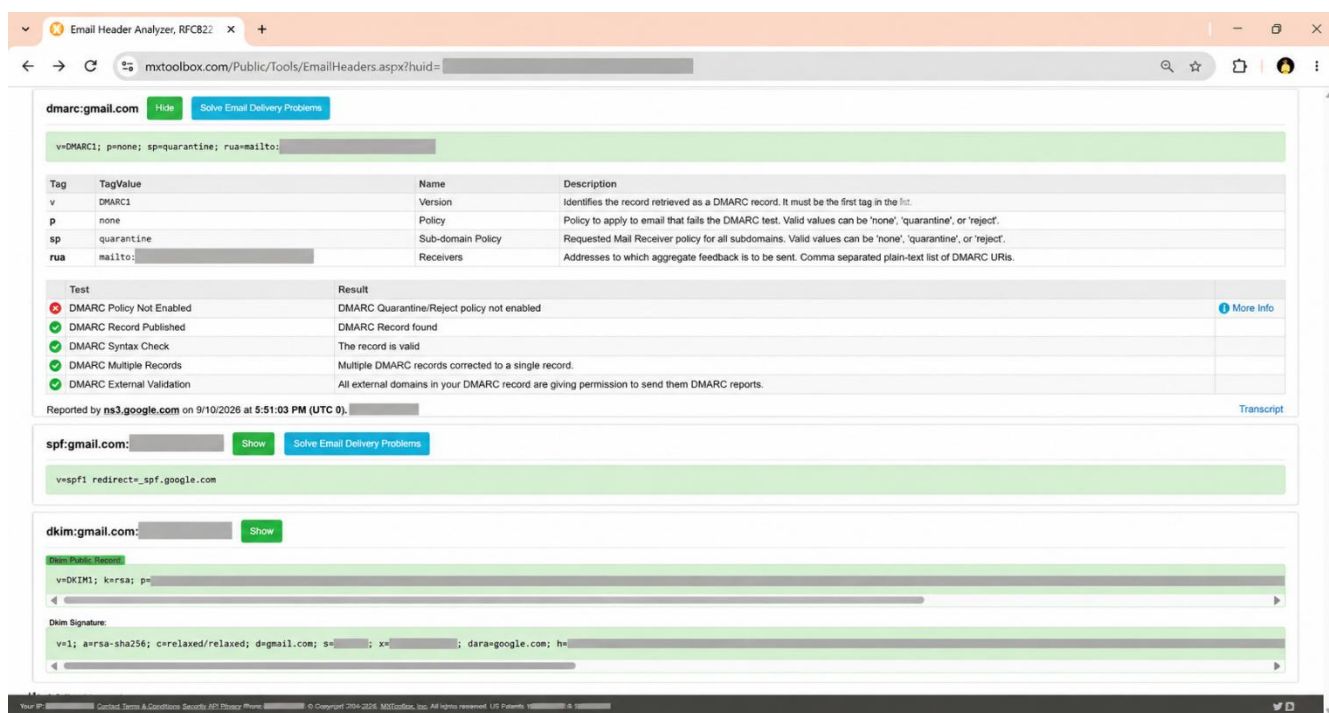


Figure 8.3 — SPF, DKIM and DMARC information examined using MXToolbox.

SPF, DKIM AND DMARC ANALYSIS

9. SPF, DKIM and DMARC Analysis

Email authentication mechanisms are important security indicators in the analysis of email messages. The project examines SPF, DKIM, and DMARC information obtained from email headers and domain records.

9.1 Sender Policy Framework (SPF)

SPF is an email authentication mechanism used to verify whether the server sending an email is authorized to send messages on behalf of a particular domain.

The SPF result obtained from the email authentication information is used as an indicator during email analysis.

Possible SPF results include:

- PASS — The sending server is authorized.
- FAIL — The sending server is not authorized.
- SOFTFAIL — The sending server is probably not authorized.
- NEUTRAL — No definitive authorization decision is provided.

9.2 DomainKeys Identified Mail (DKIM)

DKIM provides a cryptographic mechanism for validating an email message.

A DKIM signature is added to the email by the sending system. The receiving system can verify the signature using the public key published in the sender domain's DNS records.

The DKIM result is analyzed as part of the overall email-security investigation.

Possible DKIM results include:

- PASS — The DKIM signature was successfully validated.
- FAIL — The DKIM signature could not be successfully validated.
- NONE — No DKIM signature was available for verification.

9.3 Domain-based Message Authentication, Reporting and Conformance (DMARC)

DMARC builds on SPF and DKIM and provides a policy framework for handling messages that do not satisfy the domain's authentication requirements.

DMARC can provide information about:

- Authentication status
- Domain alignment

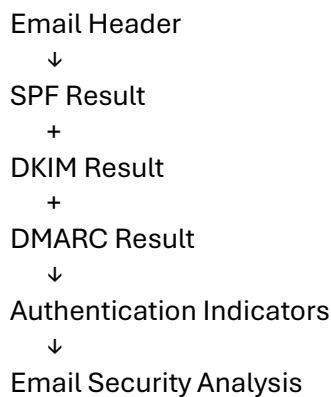
- Published domain policy
- Handling of authentication failures

The project uses DMARC information as an additional security indicator when analyzing collected email messages.

9.4 Combined Authentication Analysis

SPF, DKIM, and DMARC results are considered together rather than relying on a single authentication mechanism.

The analysis can be represented as:



Authentication results provide valuable context during investigation, but they are not sufficient by themselves to determine whether an email is malicious. They should be considered together with email content, URLs, domains, sender information, and other available security indicators.

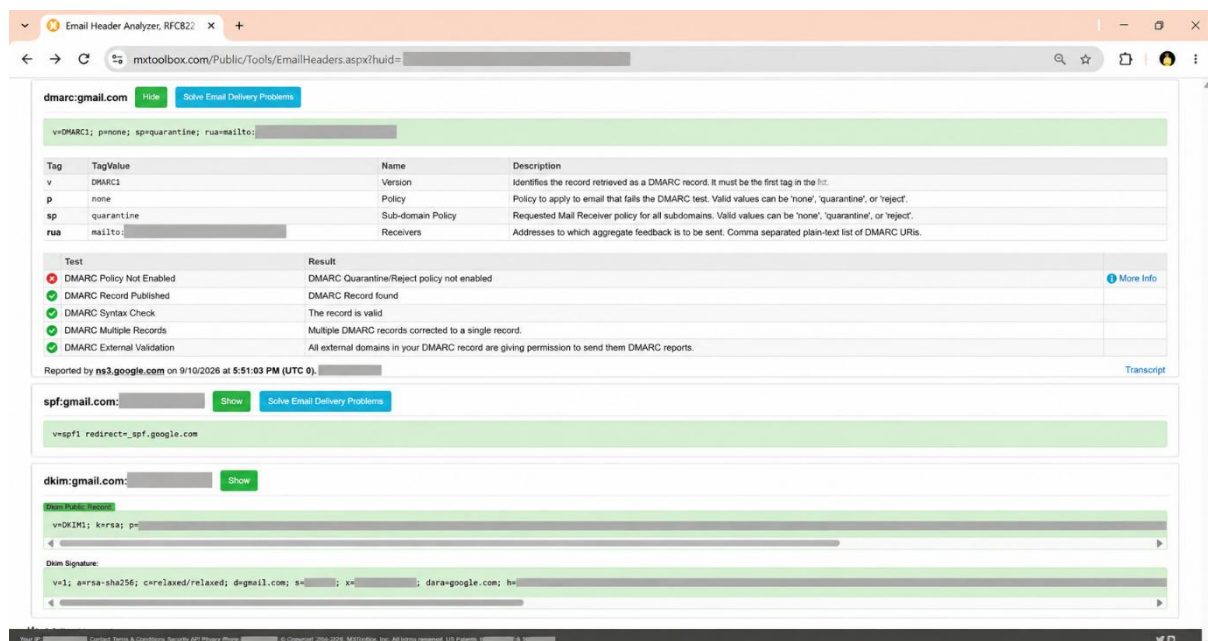


Figure 9.1 — SPF, DKIM and DMARC authentication information analyzed during the email-security investigation.

EMAIL PARSING AND STRUCTURED DATA

10. Email Parsing and Structured Data

After an email is collected through the Gmail IMAP Collector, the message is processed by the email-parsing component of the project.

The purpose of email parsing is to convert the raw email message into structured information that can be analyzed and stored for further investigation.

10.1 Information Extracted

The parser extracts important email attributes including:

- Sender
- Recipient
- Subject
- Email body
- Email headers
- URLs
- Domains
- Attachment information
- Content type
- SPF result
- DKIM result
- DMARC result

10.2 Structured Email Data

Instead of treating the complete email as a single block of text, the project separates important attributes into individual fields.

This structured format makes it easier to:

- Search specific email attributes
- Identify suspicious indicators
- Analyze authentication results
- Extract URLs and domains
- Correlate email events
- Send the information to Splunk
- Build security dashboards

10.3 Processing Workflow

Raw Email



Email Parser



Header Extraction



Body Extraction



URL & Domain Extraction



Authentication Information



Structured Email Event



Splunk

The structured event becomes the primary data source for the SIEM analysis stage.

The screenshot displays the Splunk Search & Reporting interface. At the top, a search bar contains a complex query: `host="DCB1" sourcetype="email" | eval SPF=if(match('security_headers.authentication_results','(?i)spf=pass'),'PASS','FAIL') | eval DKIM=if(match('security_headers.authentication_results','(?i)dkim=pass'),'PASS','FAIL') | eval DMARC=if(match('security_headers.authentication_results','(?i)dmarc=pass'),'PASS','FAIL')`. The search results show 5 events. The selected event is displayed in a JSON format, showing details such as the event type, mailbox, id, imap_id, sender, to, bcc, reply_to, return_path, subject, date, body, urls, attachment, content_type, dkim, and headers. The headers section includes Received, Authentication-Results, DKIM-Signature, X-Google-DKIM-Signature, ARC-Message-Signature, and ARC-Seal.

Search & Reporting

Search

host="DCB1" sourcetype="email" | eval SPF=if(match('security_headers.authentication_results','(?i)spf=pass'),'PASS','FAIL') | eval DKIM=if(match('security_headers.authentication_results','(?i)dkim=pass'),'PASS','FAIL') | eval DMARC=if(match('security_headers.authentication_results','(?i)dmarc=pass'),'PASS','FAIL')

5 events (9/9/26 10:30:00.000 PM to 9/10/26 11:23:28.000 PM) No Event Sampling

Events (5) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 50 Per Page View: Raw

SELECTED FIELDS

- host 1
- index 1
- linecount 1
- punct 3
- source 1
- sourcetype 1
- splunk_server 1

INTERESTING FIELDS

- bcc 1
- body 5
- cc 1
- content_type 3
- date 5
- DKIM 2
- DMARC 2
- domain_count 2
- domainspi 1
- event_type 1
- headers.ARC.Authentication-Results 4
- headers.ARC-Message-Signature 4
- headers.ARC-Seal 4
- headers.Authentication-Results 4
- headers.Date 2
- headers.Delivered-To 1

1 {

```
{
  "event_type": "email",
  "mailbox": "INBOX",
  "id": "<[redacted]>",
  "imap_id": "1428",
  "sender": "[redacted]@gmail.com",
  "to": "[redacted]@gmail.com",
  "bcc": "",
  "reply_to": "",
  "return_path": "[redacted]",
  "subject": "security related information",
  "date": "Thu, 10 Sep 2026 22:17:21 +0530",
  "body": "Hello,\r\n\r\nA recent security-related event was detected on your account.\r\n\r\nPlease review your recent activity ....",
  "urls": ["https://example.com/security", "https://example.com/reset"],
  "attachment": [],
  "content_type": "text/html",
  "dkim": "fail",
  "headers": {
    "Received": "from [redacted] ([redacted]) by google.com with SMTP id [redacted];",
    "Authentication-Results": "mx.google.com: spf=pass (google.com: domain of [redacted]@gmail.com designates [redacted] as permitted sender) smtp.mailfrom=[redacted]@gmail.com; dkim=fail; [redacted]@gmail.com",
    "DKIM-Signature": "v=1; a=rsa-sha256; [redacted]",
    "X-Google-DKIM-Signature": "v=1; a=rsa-sha256; [redacted]",
    "ARC-Message-Signature": "i=1; a=rsa-sha256; [redacted]",
    "ARC-Seal": "i=1; a=rsa-sha256; [redacted]"
  }
}
```

Figure 10.1 — Structured email data generated from the collected message for further security analysis and Splunk ingestion.

SPLUNK SIEM INTEGRATION

11. Splunk SIEM Integration

After the email has been collected and parsed, the structured email information is forwarded to Splunk Enterprise for centralized security monitoring and analysis.

Splunk acts as the SIEM platform for the project and provides a centralized location where collected email events can be searched, filtered, investigated, and visualized.

11.1 Email Event Ingestion

The parsed email information is stored as structured events containing relevant fields such as:

- Sender
- Recipient
- Subject
- Email body
- URLs
- Domains
- Headers
- Attachment information
- SPF result
- DKIM result
- DMARC result
- Email timestamp

These fields allow individual characteristics of an email to be analyzed without manually reviewing every message.

11.2 Splunk Event Analysis

Once the email events are available in Splunk, searches can be performed to identify specific security indicators.

Examples include:

- Searching for suspicious subjects
- Identifying emails containing URLs
- Reviewing SPF, DKIM and DMARC results
- Identifying suspicious domains
- Counting emails by sender or recipient
- Filtering suspicious email events
- Correlating multiple email indicators

11.3 Role of Splunk in the Project

Splunk provides the centralized analysis layer of the project.

The workflow is:

Gmail

↓

IMAP Collector

↓

Email Parser

↓

Structured Email Data

↓

Splunk Enterprise

↓

Search & Analysis

↓

Detection & Visualization

This integration transforms the project from a standalone email-analysis application into a SIEM-oriented security monitoring workflow.

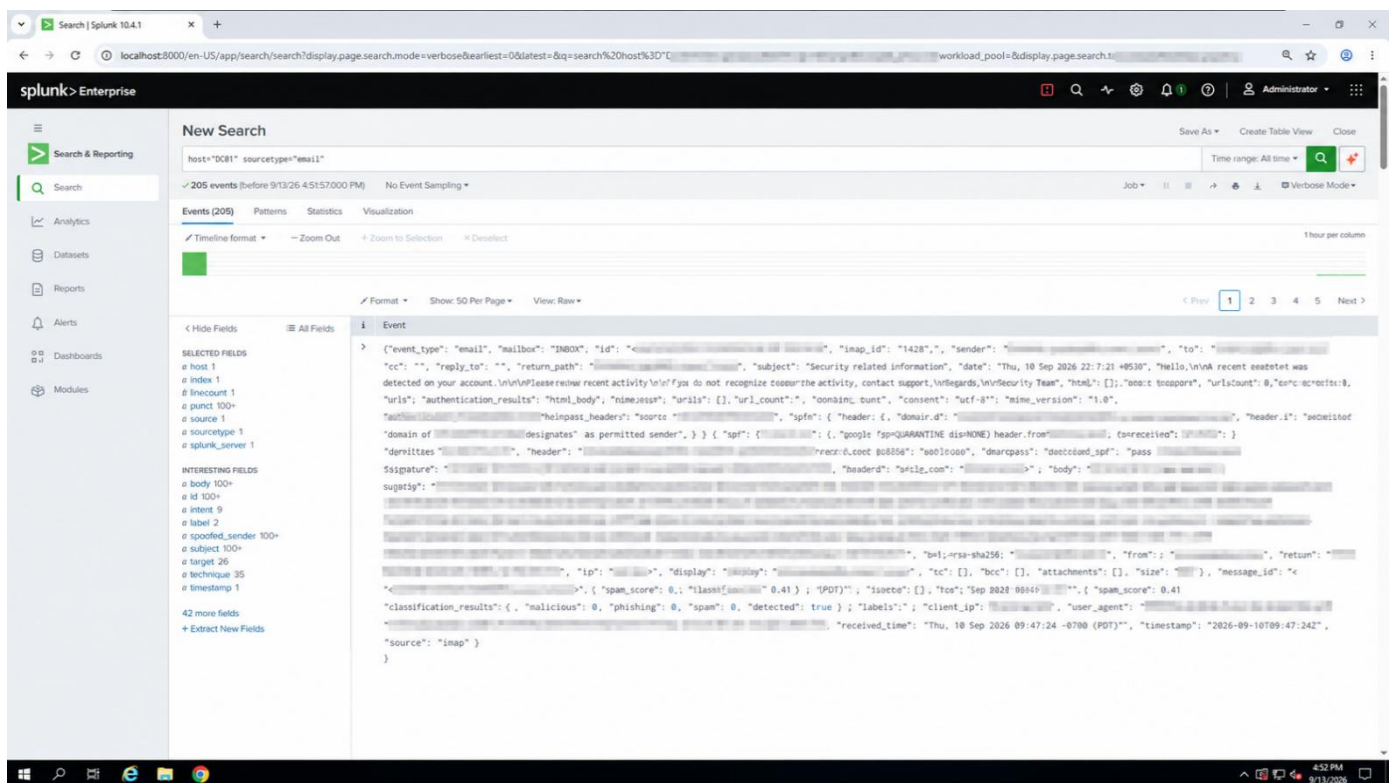


Figure 11.1 — Structured email event ingested into Splunk for centralized security analysis.

SUSPICIOUS EMAIL DETECTION

12. Suspicious Email Detection

The project performs automated detection of potentially suspicious emails using multiple indicators extracted during the email-analysis process.

12.1 Detection Indicators

An email can be marked as suspicious based on indicators such as:

- Suspicious subject patterns
- Presence of URLs
- Presence of domains
- URL and domain counts
- SPF authentication result
- DKIM authentication result
- DMARC authentication result
- Other extracted email attributes

12.2 Suspicious Email Classification

The extracted indicators are evaluated by the project's detection logic. When one or more suspicious characteristics are identified, the email is classified with an appropriate suspicious reason.

Examples of detected reasons include:

- Suspicious Subject
- Suspicious Subject + URL

This approach allows the system to provide an explanation for why an email was flagged instead of simply marking it as suspicious without context.

12.3 Splunk Detection Results

The suspicious-email results are forwarded to Splunk, where they can be searched and displayed in structured tables.

The Splunk analysis provides information such as:

- Date and time
- Sender
- Recipient
- Subject
- URL count
- Domain count
- Suspicious reason

This enables an analyst to investigate suspicious messages and understand the indicators responsible for their classification.

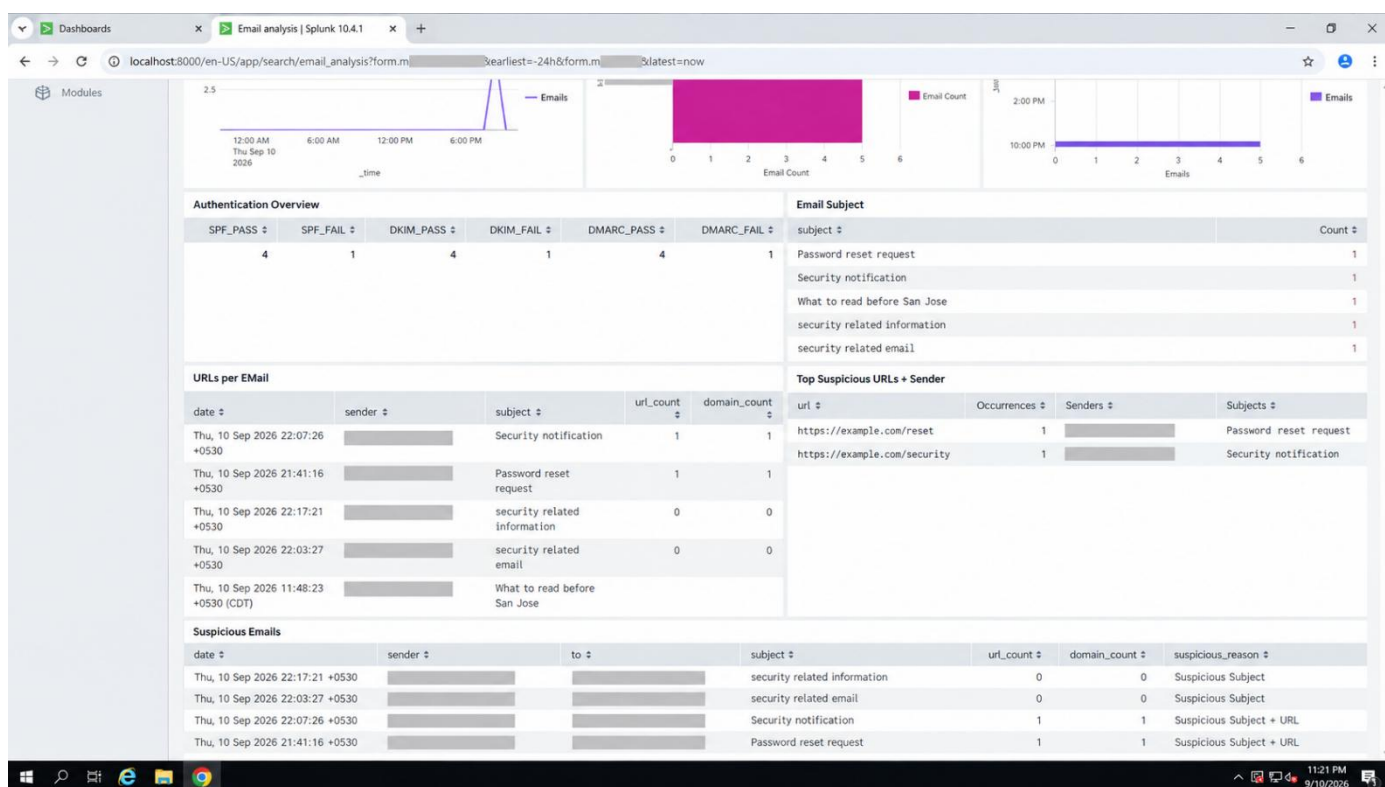


Figure 12.1 — Splunk results showing suspicious emails and the indicators responsible for their classification.

SPLUNK SECURITY DASHBOARD

13. Splunk Security Dashboard

The project uses Splunk Enterprise to provide a centralized dashboard for monitoring and analyzing collected email-security data.

The dashboard converts the processed email events into visual summaries that allow an analyst to quickly understand email activity and identify potentially suspicious messages.

13.1 Dashboard Overview

The dashboard provides an overview of the analyzed email dataset and displays key security metrics, including:

- Total number of emails
- Number of suspicious emails
- Number of emails containing domains
- Number of emails containing URLs
- Email activity over time
- Top senders
- Top recipients
- SPF results
- DKIM results
- DMARC results
- Email subjects
- URLs per email
- Top suspicious URLs

13.2 Suspicious Email Monitoring

The dashboard provides a dedicated view of suspicious email activity.

Suspicious messages can be reviewed together with their associated indicators, allowing an analyst to identify patterns and prioritize messages for further investigation.

13.3 Authentication Monitoring

SPF, DKIM, and DMARC results are visualized within the dashboard.

This allows authentication results to be reviewed alongside other email characteristics rather than analyzing each email manually.

13.4 URL and Domain Monitoring

The dashboard also provides visibility into URLs and domains extracted from email messages.

This information can be used to identify messages containing potentially suspicious links and determine which URLs require additional reputation analysis.

13.5 Test Environment Results

For the documented test dataset, the dashboard displays:

- 5 total emails
- 4 suspicious emails
- 2 emails containing domains
- 2 emails containing URLs

These values represent the specific test dataset and time range displayed in the dashboard. They should not be interpreted as the overall accuracy or detection rate of the system.

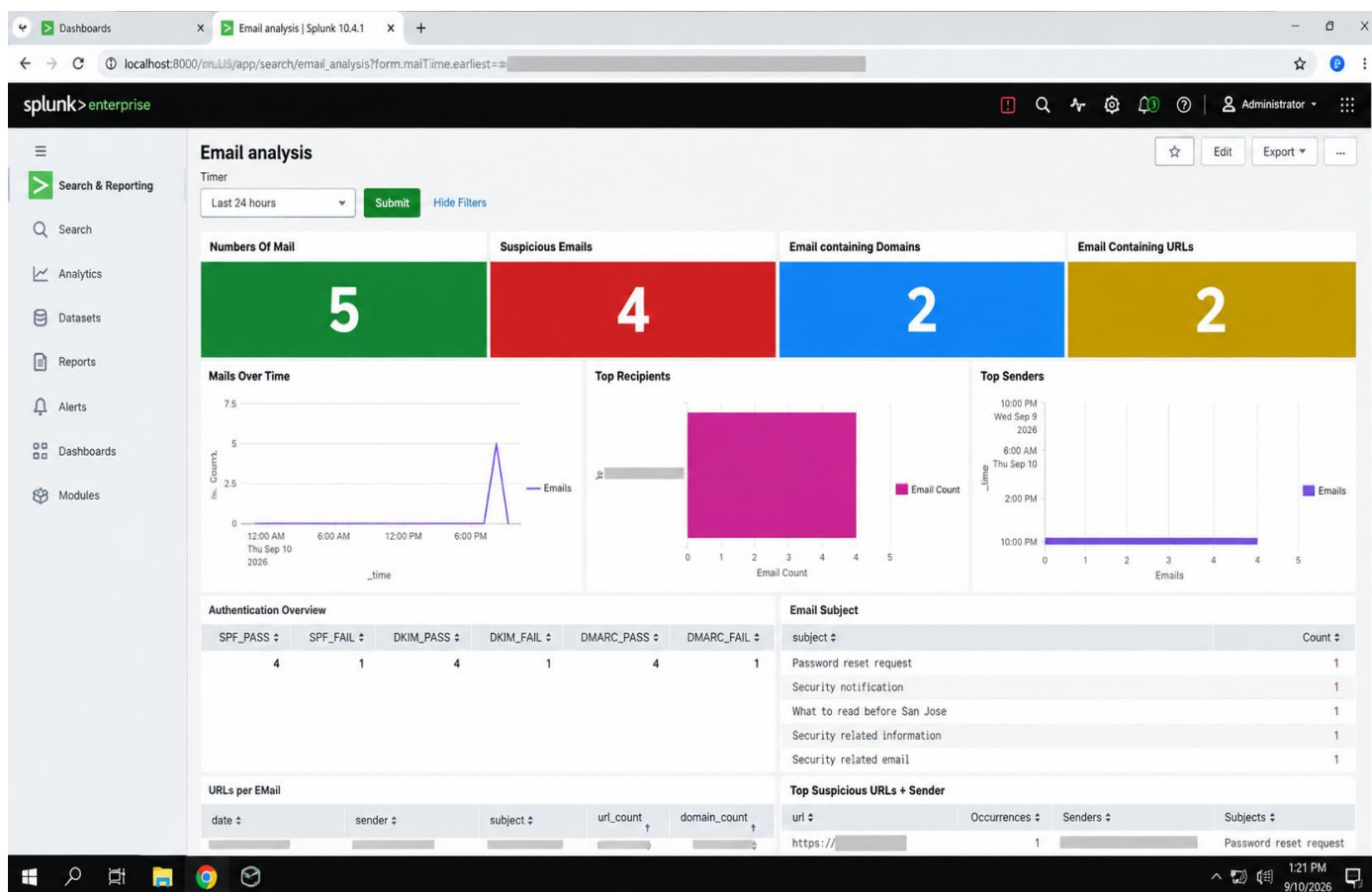


Figure 13.1 — Splunk security dashboard displaying email activity, authentication results, suspicious emails, URLs, and domains.

VIRUSTOTAL URL REPUTATION ANALYSIS

14. VirusTotal URL Reputation Analysis

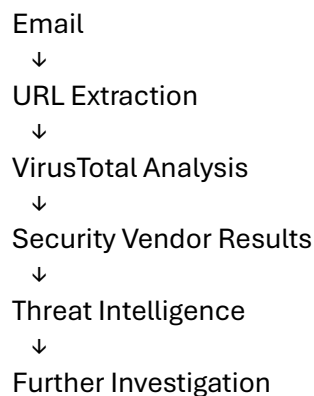
As part of the email-security analysis process, URLs extracted from email messages can be examined using VirusTotal.

The purpose of this stage is to obtain an additional threat-intelligence signal for URLs identified during email parsing.

14.1 URL Analysis Process

When a URL is extracted from an email, it can be submitted to VirusTotal for reputation analysis.

The process is:



14.2 VirusTotal Results

VirusTotal analyzes submitted URLs using multiple security vendors and services.

In the demonstrated test, the analyzed URL was reported with detections from 2 out of 90 security vendors.

This result provides additional evidence that can be considered during the investigation of the associated email.

14.3 Importance of URL Reputation

URLs are commonly used in phishing and other email-based attacks to redirect users to malicious or fraudulent websites.

URL reputation analysis can therefore provide an additional layer of information when investigating suspicious messages.

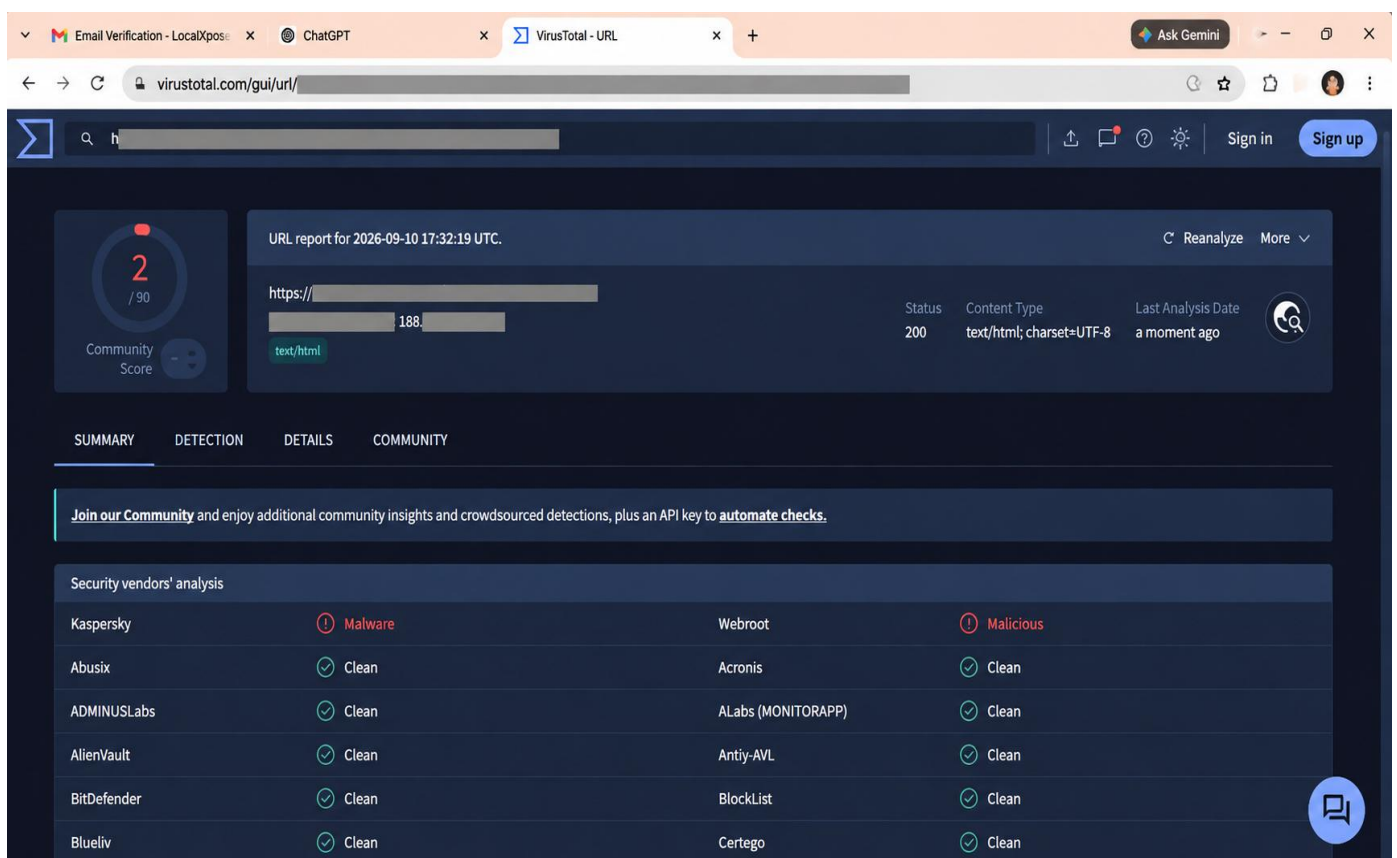
However, a VirusTotal detection result should not be treated as absolute proof that an email or URL is malicious. Reputation results should be correlated with other indicators such as:

- Email subject
- Sender information
- Domain information
- SPF result
- DKIM result
- DMARC result
- Email content
- URL characteristics

14.4 Role in the Overall Project

VirusTotal acts as an external threat-intelligence enrichment layer within the project.

The information obtained from URL analysis can support the investigation and prioritization of suspicious email events in the Splunk environment.



The screenshot displays the VirusTotal URL report interface. The top navigation bar includes links for 'Email Verification - LocalXpose', 'ChatGPT', and 'VirusTotal - URL'. The main content area shows a URL report for 2026-09-10 17:32:19 UTC. The URL is partially redacted, showing 'https://[redacted]188.[redacted]'. The status is 200, the content type is 'text/html; charset=UTF-8', and the last analysis date is 'a moment ago'. The interface also features a 'Community Score' of 2/90 and a 'Reanalyze' button. Below the summary, there is a section for 'Security vendors' analysis' with a table of results.

Security vendors' analysis	
Kaspersky	Malware
Abusix	Clean
ADMINUSLabs	Clean
AlienVault	Clean
BitDefender	Clean
Blueliv	Clean
Webroot	Malicious
Acronis	Clean
ALabs (MONITORAPP)	Clean
Antiy-AVL	Clean
BlockList	Clean
Certego	Clean

Figure 14.1 — VirusTotal analysis of a URL extracted from a test email.

TESTING AND RESULTS

15. Testing and Results

The project was tested using controlled email messages generated through the custom Gmail Mail Sender. The generated messages were delivered to the Gmail mailbox, collected through the IMAP Email Collector, parsed for security-related information, and forwarded to Splunk for analysis.

15.1 End-to-End Testing

The complete workflow was tested across the following stages:

1. Test emails were generated using the custom Gmail Mail Sender.
2. The generated messages were received in the Gmail mailbox.
3. The Gmail IMAP Collector detected and collected the messages.
4. Email content and headers were parsed into structured information.
5. SPF, DKIM, and DMARC information was extracted for analysis.
6. URLs and domains were identified from the email content.
7. Selected URLs were analyzed using VirusTotal.
8. Structured email events were indexed in Splunk.
9. Splunk searches and dashboards were used to identify and visualize suspicious email activity.

15.2 Observed Results

The documented test dataset produced the following results in the Splunk dashboard:

- Total Emails: 5
- Suspicious Emails: 4
- Emails Containing Domains: 2
- Emails Containing URLs: 2

The suspicious-email results included classifications such as:

- Suspicious Subject
- Suspicious Subject + URL

15.3 Authentication Results

The collected email events also contained SPF, DKIM, and DMARC authentication information.

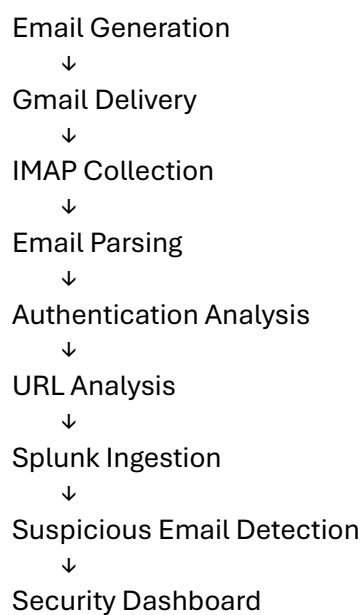
These authentication results were made available as structured fields in Splunk, allowing them to be examined together with other email indicators.

15.4 URL Analysis Results

URLs extracted from the test emails were available for further investigation. One demonstrated URL was analyzed using VirusTotal, where the displayed report showed detections from 2 out of 90 security vendors.

15.5 Test Outcome

The testing demonstrates that the project successfully connects the major stages of the email-security workflow:



The test confirms that the implemented components can work together as an end-to-end email-security analysis and SIEM monitoring pipeline.

The reported values represent the documented test dataset and should not be considered production detection accuracy or overall system performance.

SECURITY CONSIDERATIONS AND LIMITATIONS

16. Security Considerations and Limitations

The project was developed and tested in a controlled cybersecurity laboratory environment. Since the system processes email content, headers, URLs, and authentication information, appropriate security and privacy precautions must be followed.

16.1 Security Considerations

- Use a dedicated test Gmail account for project testing.
- Never store or publish Gmail passwords, app passwords, API keys, authentication tokens, or other credentials in the project repository.
- Sensitive email addresses, Message-IDs, internal IP addresses, and private email content should be removed or blurred before publishing screenshots.
- Unknown or suspicious URLs should not be opened directly on a normal personal system. They should preferably be analyzed using reputation services or an isolated analysis environment.
- Test emails should be generated only for authorized laboratory or educational purposes.
- Collected email data should be protected because it may contain sensitive information.

16.2 Detection Limitations

The current system uses observable email indicators and rule-based analysis. It should therefore be considered an email-security analysis and detection pipeline rather than a system that can definitively determine whether every email is malicious.

SPF, DKIM, and DMARC provide useful authentication information, but successful authentication does not guarantee that an email is safe.

Similarly, authentication failures do not automatically mean that an email is malicious.

VirusTotal provides an additional reputation signal, but its results may change over time and may contain false positives or false negatives.

16.3 Dataset Limitation

The documented test dataset is relatively small.

The result of 4 suspicious emails out of 5 observed emails represents only the demonstrated test scenario. It must not be presented as an 80% detection accuracy because a properly labeled dataset and ground-truth evaluation were not established.

16.4 Future Evaluation

For a production-oriented version, the system should be evaluated using a larger labeled dataset containing both legitimate and malicious emails.

Metrics such as:

- Precision
- Recall
- F1-score
- False-positive rate
- False-negative rate

could then be used to measure the effectiveness of the detection system.

Overall, the current implementation provides a practical foundation for email-security investigation while leaving scope for more advanced detection, enrichment, and automated response capabilities.

17. FUTURE IMPROVEMENTS

17.1 Machine Learning-Based Detection

The current system uses rule-based and indicator-based analysis to identify suspicious emails. A future version can integrate machine learning models trained on labeled phishing and legitimate email datasets. Features such as sender reputation, subject patterns, URL characteristics, domain information, authentication results, and email content can be used for classification.

17.2 Advanced URL and Domain Analysis

The URL analysis component can be extended with additional threat-intelligence sources and domain reputation services. Domain age, WHOIS information, DNS records, URL redirection behavior, SSL certificate information, and blacklist status can be incorporated to improve detection.

17.3 Automated Threat Scoring

A centralized risk-scoring mechanism can be implemented to assign a numerical risk score to every email. Different indicators such as suspicious subjects, malicious URLs, domain reputation, and authentication failures can contribute to the final score.

17.4 Improved Splunk Detection and Alerting

Additional Splunk correlation searches and detection rules can be created for repeated phishing attempts, suspicious sender behavior, authentication failures, malicious URLs, and unusual email activity. Automated alerts can notify security analysts when predefined risk thresholds are reached.

17.5 Larger Dataset and Model Evaluation

Future testing should use a larger and properly labeled dataset containing both legitimate and phishing emails. Performance can be evaluated using precision, recall, F1-score, false-positive rate, and false-negative rate to provide a more reliable assessment of detection performance.

17.6 Automated Incident Response

The pipeline can be extended with automated response capabilities. High-risk emails could be flagged, quarantined, or forwarded for analyst review based on predefined security policies.

17.7 Enhanced Dashboard

The Splunk dashboard can be expanded with additional visualizations such as phishing trends, top suspicious domains, frequently detected URLs, authentication failure trends, sender reputation, risk-score distribution, and time-based attack patterns.

17.8 Attachment Analysis

Future versions can analyze email attachments for suspicious file types, hashes, embedded URLs, macros, and other indicators of compromise. File hashes can also be checked against external threat-intelligence services.

17.9 Scalability and Real-Time Processing

The system can be optimized for continuous monitoring and larger email volumes. Improved processing, caching, asynchronous collection, and real-time Splunk ingestion can make the pipeline more suitable for a SOC environment.

17.10 Overall Future Direction

The long-term goal is to develop the current prototype into a more comprehensive email security monitoring platform that combines automated email collection, authentication analysis, threat intelligence, intelligent detection, SIEM monitoring, risk scoring, alerting, and incident-response capabilities.

18. CONCLUSION

The Email Security Analysis & Phishing Detection Pipeline successfully demonstrates an end-to-end approach for collecting, analyzing, enriching, and monitoring email security data.

The project integrates a custom Gmail Mail Sender, Gmail IMAP collection, email parsing, header analysis, SPF/DKIM/DMARC information extraction, URL and domain analysis, VirusTotal enrichment, and Splunk Enterprise for centralized security monitoring.

The developed pipeline converts raw email messages into structured security information that can be searched and analyzed within a SIEM environment. Suspicious email indicators such as suspicious subjects, URLs, domains, and authentication results can be used to support security investigations and detection activities.

Testing demonstrated that the complete workflow operates successfully from controlled email generation through Gmail collection, analysis, VirusTotal enrichment, and Splunk visualization. The test dataset produced observable suspicious-email classifications and corresponding security indicators.

Although the current implementation is primarily rule-based and was evaluated using a limited test dataset, it provides a strong foundation for further development. Machine learning, advanced threat intelligence, automated alerting, risk scoring, attachment analysis, and automated incident response can be incorporated to improve the system.

Overall, this project demonstrates practical application of cybersecurity concepts including email security, phishing detection, digital investigation, threat intelligence, and SIEM-based monitoring. It also provides a foundation for developing a more advanced email security monitoring solution suitable for SOC-oriented environments